

120 DAYS OF QUANT FINANCE

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INTRODUCTION TO PROBABILITY FOR QUANT FINANCE

Probability theory is not merely a collection of formulas—it is a modeling language for uncertainty.

In quantitative disciplines, particularly finance, we are rarely interested in deterministic outcomes. Instead, we model distributions of possible outcomes, quantify uncertainty, and reason about events that have not yet occurred. Probability provides the formal structure to do exactly this.

In this series, probability will be treated not as an abstract academic subject, but as the foundation upon which stochastic models, simulations, and financial pricing frameworks are built. Every model we use later—whether for asset returns, volatility, or derivatives—will ultimately rely on probabilistic assumptions introduced here.

This first module focuses purely on the mathematical foundations: sample spaces, events, probability measures, and conditional reasoning. While financial applications are intentionally deferred, the concepts developed here will be directly translated into code and financial models in subsequent days.

A precise understanding of probability is non-negotiable for any aspiring quantitative researcher. Errors at this level propagate silently—and expensively—into real-world models.

SAMPLE SPACE

- In probability theory, the sample space (also called sample description space, the possibility space, or outcome space) of an experiment or random trial is the set of all possible outcomes or results of that experiment. A sample space is usually denoted using the set notation, and the possible ordered outcomes, or sample points, are listed as elements in the set.

It is common to refer to a sample space by the labels S , Ω , or U (for “universal set”). The elements of a sample space may be numbers, words, letters, or symbols. They can also be finite, countably infinite, or uncountably infinite.

A subset of the sample space is an event, denoted by E . If the outcome of an experiment is included in E , then event E has occurred.

- **Types of Sample Spaces:**

There are three types of Sample space:

1. **Finite:** The number of outcomes is countable and limited (e.g., rolling a die). Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$
2. **Infinite:** The outcomes go on forever (e.g., the Number of coin tosses needed until the first head appears). Sample Space: $S = \{1, 2, 3, 4, 5, 6, \dots\}$
3. **Continuous:** The outcomes can take any real value within a given range (e.g., A real number between 0 and 1). Sample space: $S = \{x \in R \mid 0 \leq x \leq 1\}$

DEFINITION OF PROBABILITY

- Probability is a branch of mathematics and statistics concerning events and numerical descriptions of how likely they are to occur. The probability of an event is a number between 0 and 1; the larger the probability, the more likely an event is to occur. A simple example is the tossing of a fair (unbiased) coin. Since the coin is fair, the two outcomes ("heads" and "tails") are both equally probable; the probability of "heads" equals the probability of "tails"; and since no other outcomes are possible, the probability of either "heads" or "tails" is 1/2 (which could also be written as 0.5).

In modern probability theory, probability is defined axiomatically as a measure P that assigns a number in $[0,1]$ to events in a sample space Ω , satisfying the Kolmogorov axioms.

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

MUTUALLY EXCLUSIVE EVENTS

Mutually exclusive events are two or more events that cannot occur at the same time or simultaneously in a single trial of a random experiment. If one event occurs, the other cannot occur, meaning their joint probability is zero. This is mathematically expressed as $P(A \cap B) = 0$, where A and B are mutually exclusive events. It is important to note that mutually exclusive events are not independent. In fact, they are dependent by definition—knowing that one event has occurred provides certainty that the other did not occur.

INCLUSIVE-EXCLUSIVE PRINCIPLE

In combinatorics, the inclusion–exclusion principle is a counting technique which generalizes the familiar method of obtaining the number of elements in the union of two finite sets; symbolically expressed as:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P\left(\bigcap_{i=1}^n A_i\right)$$

AXIOMS OF PROBABILITY

The standard probability axioms are the foundations of probability theory introduced by The Russian mathematician Andrey Kolmogorov in 1933. Like all axiomatic systems, they outline the basic assumptions underlying the application of probability to fields such as pure mathematics and the Physical sciences, while avoiding logical paradoxes.

In order to state the Kolmogorov axioms, the following pieces of data must be specified:

1. The sample space, Ω , which is the set of all possible outcomes or elementary events.
2. The space of all events, which are each taken to be sets of outcomes (i.e. subsets of Ω).
3. The probability measure P which assigns to each event $E \in F$ its probability, $P(E)$.

Here are the Three AXIOMS OF PROBABILITY

- AXIOM 1:

The probability of an event E is a non-negative real number. This assumption is implied by the fact that P is a measure on F .

$$P(E) \geq 0 \quad \forall E \in F$$

- AXIOM 2:

This is the assumption of unit measure: that the probability that at least one of the elementary events in the entire sample space will occur is 1.

$$P(\Omega) = 1$$

From this axiom it follows that $P(E)$ is always finite, in contrast with more general measure theory.

- AXIOM 3: Any countable sequence of disjoint sets (synonymous with mutually exclusive events) $E_1, E_2 \dots$ satisfies

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

This property again is implied by the fact that P is a measure. Note that, by taking $E_1 = \Omega$ and $E_i = \emptyset$ for all $i > 1$, one deduces that $P(\emptyset) = 0$.

ELEMENTARY CONSEQUENCES OF AXIOMS OF PROBABILITY

In order to demonstrate that the theory generated by the Kolmogorov axioms corresponds with classical probability, some elementary consequences are typically derived:

1. COMPLEMENT RULE

Since P is Countably additive, we have $P(A) + P(A^c) = P(A \cup A^c) = P(\Omega) = 1$,
So,

$$P(A^c) = 1 - P(A)$$

2. PROBABILITY RULE

In particular, it follows that $P(\emptyset) = 0$. The empty set is interpreted as the event that "no outcome occurs", which is impossible.

3. MONOTONICITY

Similarly, if $A \subseteq B$, then $P(B) = P(A \cup (B \setminus A)) = P(A) + P(B \setminus A) \geq P(A)$.
In other words, P is monotone.

4. BOUNDED PROBABILITY

Since $\emptyset \subseteq E \subseteq \Omega$ for any event E , it follows that $0 \leq P(E) \leq 1$.

By dividing $A \cup B$ into the disjoint sets $A \setminus (A \cap B)$, $B \setminus (A \cap B)$ and $A \cap B$, one arrives at a probabilistic version of the inclusion-exclusion principle

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

In the case where Ω is finite, the two identities are equivalent.

PROBABILITY SPACE

In probability theory, a probability space or a probability triple $(\Omega, \mathcal{F}, P)$ is a mathematical construct that provides a formal model of a random process or "experiment".

- A probability space consists of three elements: A sample space, Ω , which is the set of all possible outcomes of a random process under consideration.
- An event space, $\mathcal{F}$, which is a set of events, where an event is a subset of outcomes in the sample space.
- A probability function, P , which assigns, to each event in the event space, a probability, which is a number between 0 and 1 (inclusive).

In order to provide a model of probability, these elements must satisfy probability axioms.

INDEPENDENCE

Two events are said to be independent if knowing the outcome of one event does not change your belief about whether or not the other event will occur. For example, you might say that two separate dice rolls are independent of one another: the outcome of the first dice gives you no information about the outcome of the second dice and vice versa.

$$P(A \cap B) = P(A)P(B)$$

CONDITIONAL PROBABILITY

In probability theory, conditional probability is a measure of the probability of an event occurring, given that another event (by assumption, presumption, assertion or evidence) is already known to have occurred. This particular method relies on event A occurring with some sort of relationship with another event B. In this situation, the event A can be analyzed by a conditional probability with respect to B. If the event of interest is A and the event B is known or assumed to have occurred, "the conditional probability of A given B", or "the probability of A under the condition B", is usually written as $P(A | B)$. This can also be understood as the fraction of probability B that intersects with A, or the ratio of the probabilities of both events happening to the "given" one happening (how many times A occurs rather than not assuming B has occurred).

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

LAW OF TOTAL PROBABILITY

The law of total probability is a theorem that states, in its discrete case, if $\{B_n : n = 1, 2, 3, \dots\}$ is a finite or countably infinite set of mutually exclusive and collectively exhaustive events, then for any event A :

$$P(A) = \sum_n P(A | B_n)P(B_n),$$

where, for any n , if $P(B_n) = 0$, then these terms are simply omitted from the summation since $P(A | B_n)$ is finite.

BAYES' THEOREM

Bayes' Theorem is a mathematical formula used to determine the conditional probability of an event based on prior knowledge of conditions that might be related to the event. It provides a way to update the probability of a hypothesis as more evidence or information becomes available. Bayes' theorem is stated mathematically as the following equation:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where A and B are events and $P(B) \neq 0$. $P(A|B)$ is a conditional probability: the probability of event A occurring given that B is true. It is also called the posterior probability of A given B . $P(B|A)$ is also a conditional probability: the probability of event B occurring given that A is true. It can also be interpreted as the likelihood of A given a fixed B . $P(A)$ and $P(B)$ are the probabilities of observing A and B respectively without any given conditions; they are known as the prior probability and marginal probability/Normalization constant.

There are several techniques for handling the case where the denominator is not known. One technique is to use the law of total probability to expand out the term, resulting in another formula, called Bayes' Theorem with Law of Total Probability:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}.$$

For proposition A and evidence or background B , $P(A)$ is the prior probability, the initial degree of belief in $P(A^c)$ is the corresponding initial degree of belief in not- A , that A is false, where $P(A^c) = 1 - P(A)$. $P(B|A)$ is the conditional probability or likelihood, the degree of belief in B given that A is true. $P(B|A^c)$ is the conditional probability or likelihood, the degree of belief in B given that A is false. $P(A|B)$ is the posterior probability, the probability of A after taking into account B .

EXAMPLES

Now We will look at some examples from each of the topic.

1. Finite Sample Space

X_1 = result of the first roll, and X_2 = result of the second roll.
Write down the sample space S , and assuming that all outcomes are equally likely (because the die is fair), find the probability of the event A defined as the event that: $X_1 + X_2 = 8$ I roll a fair die twice and obtain two numbers:

X_1 = result of the first roll, and X_2 = result of the second roll.
Write down the sample space S , and assuming that all outcomes are equally likely (because the die is fair), find the probability of the event A defined as the event that: $X_1 + X_2 = 8$

SOLUTION:

The sample space S can be written as $S =$

$(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),$
 $(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),$
 $(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),$
 $(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),$
 $(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),$
 $(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)$.

As we see there are $|S| = 36$ elements in S . To find probability of A , all we need to do is find $M = |A|$. In particular, A is defined as

$$A = \{(X_1, X_2) \mid X_1 + X_2 = 8, X_1, X_2 \in \{1, 2, \dots, 6\}\} = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$$

Thus, $|A| = 5$, which means that:

$$P(A) = \frac{|A|}{|S|} = \frac{5}{36}$$

2. Conditional Probability Application

Celine is undecided as to whether to take a French course or a chemistry course. She estimates that her probability of receiving an A grade would be $\frac{1}{2}$ in a French course and $\frac{2}{3}$ in a chemistry course. If Celine decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?

SOLUTION:

Let the event that Celine takes chemistry and A denote the event that she receives an A in whatever course she takes, then the desired probability is $P(CA)$, which is calculated as:

$$\begin{aligned}
 P(CA) &= P(C)P(A|C) \\
 &= (1/2)(2/3) \\
 &= 1/3
 \end{aligned}$$

3. Testing Independence

I pick a random number from $\{1, 2, 3, \dots, 10\}$, and call it N . Suppose that all outcomes are equally likely. Let A be the event that N is less than 7, and let B be the event that N is an even number. Are A and B independent?

SOLUTION:

We have $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8, 10\}$, and $A \cap B = \{2, 4, 6\}$. Then,

$$P(A) = 0.6$$

$$P(B) = 0.5$$

$$P(A \cap B) = 0.3$$

Therefore, $P(A \cap B) = P(A)P(B)$, so A and B are independent. This means that knowing that B has occurred does not change our belief about the probability of A . In this problem the two events are about the same random number, but they are still independent because they satisfy the definition.

4. Law of Total Probability

I have three bags that each contain 100 marbles:

- (a) Bag 1 has 75 red and 25 blue marbles;
- (b) Bag 2 has 60 red and 40 blue marbles;
- (c) Bag 3 has 45 red and 55 blue marbles.

I choose one of the bags at random and then pick a marble from the chosen bag, also at random. What is the probability that the chosen marble is red?

SOLUTION:

Let R be the event that the chosen marble is red. Let B_i be the event that I choose Bag i . We already know that

$$P(R|B_1) = 0.75,$$

$$P(R|B_2) = 0.60,$$

$$P(R|B_3) = 0.45$$

We choose our partition as B_1, B_2, B_3 . Note that this is a valid partition because, firstly, the B_i 's are disjoint (only one of them can happen), and secondly, because their union

is the entire sample space as one of the bags will be chosen for sure, i.e., $P(B_1 \cup B_2 \cup B_3) = 1$. Using the law of total probability, we can write

$$P(R) = P(R|B_1)P(B_1) + P(R|B_2)P(B_2) + P(R|B_3)P(B_3)$$

$$P(R) = (0.75)\frac{1}{3} + (0.60)\frac{1}{3} + (0.45)\frac{1}{3}$$

$$P(R) = 0.60$$

5. Bayes' Theorem (Medical Testing / Base Rate Fallacy)

If all patients with pancreatic cancer have a certain symptom, it does not follow that anyone who has that symptom has a 100% chance of getting pancreatic cancer. Assuming the incidence rate of pancreatic cancer is 1/100000, while 10/99999 healthy individuals have the same symptoms worldwide, the probability of having pancreatic cancer given the symptoms is 9.1%, and the other 90.9% could be "false positives" (that is, falsely said to have cancer; "positive" is a confusing term when, as here, the test gives bad news).

$$P(Cancer|Symptomes) = \frac{P(Symptomes|Cancer)P(Cancer)}{P(Symptomes)}$$

$$= \frac{P(Symptomes|Cancer)P(Cancer)}{P(Symptomes|Cancer)P(Cancer) + P(Symptomes|Non - Cancer)P(Non - Cancer)}$$

$$= \frac{1 * 0.00001}{1 * 0.00001 + \frac{10}{99999} * 0.99999}$$

$$= \frac{1}{11} = 0.0909.....$$

SUMMARY

Today we covered:

1. Sample spaces (finite, countably infinite, continuous)
2. Three interpretations of probability
3. Kolmogorov's axiomatic framework
4. Elementary consequences and properties
5. Mutually exclusive vs. independent events
6. Conditional probability
7. Law of total probability
8. Bayes' Theorem

Next (Day 2): Random variables, probability distributions, expectation, variance, and moment generating functions.

RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

In the previous module, probability was introduced as a mathematical framework for reasoning about uncertainty through events and probability measures. In practice, however, we are rarely interested in events alone. Instead, we study numerical quantities derived from random experiments—quantities whose values depend on the underlying randomness.

A random variable formalizes this idea by assigning a real number to each outcome in a sample space. Once a random variable is defined, the central object of interest becomes its probability distribution, which describes how probability mass or density is distributed across possible values.

This module focuses on the mathematical structure of probability distributions, distinguishing between discrete and continuous cases. We then introduce summary measures—expectation, variance, skewness, and kurtosis—which capture different aspects of a distribution’s shape and behavior.

These concepts are developed here in a purely mathematical setting. Their role as foundational building blocks for stochastic modeling and later applications will become evident as we proceed.

Random Variables

A **random variable** is a measurable function

$$X : \Omega \rightarrow R,$$

which assigns a real number to each outcome in the sample space Ω .

Discrete Probability Distributions

A random variable X is said to be **discrete** if it takes values in a finite or countably infinite set.

The probability distribution of a discrete random variable is described by its **probability mass function (PMF)**:

$$p_X(x) = P(X = x),$$

where x belongs to the support of X . The PMF satisfies:

$$p_X(x) \geq 0 \quad \text{and} \quad \sum_x p_X(x) = 1.$$

Continuous Probability Distributions

A random variable X is said to be **continuous** if there exists a **probability density function (PDF)** $f_X(x)$ such that for any interval $[a, b]$,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

The PDF satisfies: $f_X(x) \geq 0$ and $\int_{-\infty}^{\infty} f_X(x) dx = 1$.
For a continuous random variable,

$$P(X = x) = 0$$

for any real number x .

Probability Mass Functions, Density Functions, and Distribution Functions

To fully describe the distribution of a random variable, it is often useful to work with functions that encode how probability is assigned to different values. These include probability mass functions, probability density functions, and cumulative distribution functions.

Probability Mass Function (PMF)

For a discrete random variable X , the **probability mass function (PMF)** is defined as:

$$p_X(x) = P(X = x).$$

The PMF satisfies:

$$p_X(x) \geq 0 \quad \text{and} \quad \sum_x p_X(x) = 1.$$

The PMF completely characterizes the distribution of a discrete random variable.

Probability Density Function (PDF)

For a continuous random variable X , the distribution is described by a **probability density function (PDF)** $f_X(x)$ such that:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

The PDF satisfies:

$$f_X(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

It is important to note that the value of the PDF at a point is not a probability:

$$P(X = x) = 0 \quad \text{for continuous random variables.}$$

Cumulative Distribution Function (CDF)

The **cumulative distribution function (CDF)** of a random variable X is defined as:

$$F_X(x) = P(X \leq x).$$

The CDF exists for both discrete and continuous random variables and uniquely determines the distribution of X .

Properties of the CDF

The cumulative distribution function has the following properties:

- $0 \leq F_X(x) \leq 1$ for all $x \in R$.
- $F_X(x)$ is non-decreasing.

- $\lim_{x \rightarrow -\infty} F_X(x) = 0.$
- $\lim_{x \rightarrow \infty} F_X(x) = 1.$

For a continuous random variable with PDF $f_X(x)$,

$$F_X(x) = \int_{-\infty}^x f_X(t) dt,$$

and

$$f_X(x) = \frac{d}{dx} F_X(x),$$

wherever the derivative exists.

Important Probability Distributions

Certain probability distributions appear repeatedly across probability theory due to their mathematical properties and interpretability. We introduce some of the most fundamental distributions below.

Bernoulli Distribution

A random variable X is said to follow a **Bernoulli distribution** with parameter $p \in [0, 1]$ if:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

The probability mass function is:

$$p_X(x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0. \end{cases}$$

The expectation and variance are given by:

$$E[X] = p, \quad \text{Var}(X) = p(1 - p).$$

Binomial Distribution

A random variable X follows a **Binomial distribution** with parameters $n \in \mathbb{N}$ and $p \in [0, 1]$ if it represents the number of successes in n independent Bernoulli trials, each with success probability p .

The probability mass function is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

The expectation and variance are:

$$E[X] = np, \quad \text{Var}(X) = np(1 - p).$$

Poisson Distribution

A random variable X is said to follow a **Poisson distribution** with parameter $\lambda > 0$ if its probability mass function is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

The expectation and variance are both equal to:

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

The Poisson distribution commonly arises as a limit of the Binomial distribution when the number of trials is large and the success probability is small.

Normal Distribution

A continuous random variable X is said to follow a **Normal distribution** with mean $\mu \in R$ and variance $\sigma^2 > 0$ if its probability density function is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in R.$$

The expectation and variance are:

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

The Normal distribution is symmetric about its mean, with skewness equal to zero and kurtosis equal to three.

Expectation

The **expectation** (or mean) of a random variable X , denoted by $E[X]$, is a weighted average of its possible values.

For a discrete random variable:

$$E[X] = \sum_x x P(X = x).$$

For a continuous random variable:

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx,$$

provided the integral exists.

Expectation is a linear operator. For constants a, b and random variables X, Y ,

$$E[aX + bY] = aE[X] + bE[Y],$$

whenever the expectations exist.

Variance

The **variance** of a random variable X , denoted by $\text{Var}(X)$, measures the dispersion of the distribution around its mean.

It is defined as:

$$\text{Var}(X) = E[(X - E[X])^2].$$

An equivalent expression is:

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

For constants a, b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Skewness

Skewness measures the asymmetry of a probability distribution around its mean.

It is defined as the third standardized moment:

$$\text{Skewness}(X) = \frac{E[(X - \mu)^3]}{\sigma^3},$$

where $\mu = E[X]$ and $\sigma = \sqrt{\text{Var}(X)}$.

A positive skew indicates a longer right tail, while a negative skew indicates a longer left tail.

Kurtosis

Kurtosis measures the heaviness of the tails of a distribution relative to its center.

It is defined as the fourth standardized moment:

$$\text{Kurtosis}(X) = \frac{E[(X - \mu)^4]}{\sigma^4}.$$

The **excess kurtosis** is defined as:

$$\text{Excess Kurtosis} = \text{Kurtosis} - 3,$$

so that a normal distribution has excess kurtosis equal to zero.

Examples

Example 1: Discrete Random Variable

Consider a fair six-sided die rolled once. Let the random variable X denote the number obtained on the top face.

The sample space is:

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Since the die is fair, each outcome is equally likely, and the probability mass function is given by:

$$P(X = x) = \frac{1}{6}, \quad x \in \{1, 2, 3, 4, 5, 6\}.$$

The expectation of X is:

$$E[X] = \sum_{x=1}^6 x \cdot \frac{1}{6} = \frac{21}{6} = 3.5.$$

Example 2: Continuous Random Variable

Let X be a continuous random variable uniformly distributed on the interval $[0, 1]$. Its probability density function is:

$$f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The expectation of X is:

$$E[X] = \int_0^1 x \, dx = \frac{1}{2}.$$

The variance of X is:

$$\text{Var}(X) = \int_0^1 (x - \frac{1}{2})^2 \, dx = \frac{1}{12}.$$

Example 3: Expectation of a Function of a Random Variable

Let X be a discrete random variable with probability mass function:

$$P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{2}.$$

The expectation of X^2 is:

$$E[X^2] = 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{2} = \frac{5}{2}.$$

This illustrates that, in general,

$$E[X^2] \neq (E[X])^2.$$

Example 4: Variance Using the Alternative Formula

Using the same random variable as in Example 3, we compute:

$$E[X] = \frac{1+2}{2} = \frac{3}{2},$$
$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{5}{2} - \left(\frac{3}{2}\right)^2 = \frac{1}{4}.$$

This example demonstrates the convenience of the identity $\text{Var}(X) = E[X^2] - (E[X])^2$.

Example 5: Skewness and Symmetry

Let X be a continuous random variable whose probability density function is symmetric about its mean μ . Then:

$$E[(X - \mu)^3] = 0,$$

and therefore the skewness of X is zero.

This shows that skewness captures asymmetry in a distribution rather than its spread.

Examples: Important Probability Distributions

Example 6: Bernoulli Distribution

Let X be a Bernoulli random variable with parameter $p = 0.3$. That is,

$$P(X = 1) = 0.3, \quad P(X = 0) = 0.7.$$

The expectation of X is:

$$E[X] = p = 0.3,$$

and the variance is:

$$\text{Var}(X) = p(1 - p) = 0.21.$$

This example illustrates the simplest discrete random variable, taking only two possible values.

Example 7: Binomial Distribution

Let X follow a Binomial distribution with parameters $n = 10$ and $p = 0.4$. Then X represents the number of successes in 10 independent Bernoulli trials.

The probability that exactly four successes occur is:

$$P(X = 4) = \binom{10}{4} (0.4)^4 (0.6)^6.$$

The expectation and variance are:

$$E[X] = np = 4,$$

$$\text{Var}(X) = np(1 - p) = 2.4.$$

Example 8: Poisson Distribution

Let X follow a Poisson distribution with parameter $\lambda = 3$.

The probability that X takes the value 2 is:

$$P(X = 2) = \frac{3^2 e^{-3}}{2!}.$$

The expectation and variance of X are both equal to:

$$E[X] = \text{Var}(X) = 3.$$

This example highlights the defining property of the Poisson distribution: its mean and variance are equal.

Example 9: Normal Distribution

Let X follow a Normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$.

The probability that X lies within one standard deviation of the mean is:

$$P(-1 \leq X \leq 1) = \int_{-1}^1 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

Although this integral has no closed-form expression, its numerical value is approximately:

$$P(-1 \leq X \leq 1) \approx 0.68.$$

SUMMARY

Today we covered:

1. Random Variables
2. Discrete/Continuous Probability Distributions
3. Probability Distribution Function(PDF)
4. Probability Mass Function(PMF)
5. Cumulative Distribution Function(CMF)
6. Important Probability Distributions
7. Expectation
8. Variance
9. Skewness
10. Kurtosis

Next (Day 3): Implementation of distributions, Visualizing PDFs, CDFs, Simulate Coin flip, dice rolls, Poisson events

Introduction

In previous modules, probability distributions were introduced in an abstract mathematical setting. In this module, we make the first explicit transition from pure probability theory to financial modeling.

Financial models are not direct descriptions of reality; they are structured approximations built on assumptions. One of the most classical and influential assumptions in quantitative finance is that asset returns follow a Normal distribution. While this assumption is known to be imperfect, it provides a mathematically tractable framework for pricing, risk measurement, and scenario generation.

This module introduces the Normal distribution as a model for asset returns and demonstrates how this assumption leads naturally to a quantile-based risk measure known as Value at Risk (VaR).

Modeling Asset Returns

Let P_t denote the price of an asset at time t . We define the **logarithmic return** between times $t - 1$ and t as:

$$R_t = \ln \left(\frac{P_t}{P_{t-1}} \right).$$

Log-returns are commonly used due to their additive property over time and their compatibility with continuous-time models.

Normality Assumption

The classical modeling assumption is that asset returns are Normally distributed:

$$R_t \sim \mathcal{N}(\mu, \sigma^2),$$

where:

- μ represents the expected return,
- σ represents the volatility of returns.

This assumption implies that returns are symmetric, have finite moments of all orders, and are fully characterized by their mean and variance.

It is important to emphasize that this assumption is made for analytical convenience and does not claim to perfectly describe empirical return distributions.

From Returns to Price Distributions

If log-returns are Normally distributed, then asset prices follow a **lognormal distribution**. Specifically, over a time horizon T ,

$$P_T = P_0 \exp \left(\sum_{t=1}^T R_t \right).$$

Under the assumption of independent and identically distributed returns,

$$\sum_{t=1}^T R_t \sim \mathcal{N}(T\mu, T\sigma^2).$$

This result implies that:

- Expected returns scale linearly with time,
- Variance scales linearly with time,
- Prices remain strictly positive.

These properties form the mathematical backbone of many classical asset pricing models.

Quantiles of the Return Distribution

Given a random variable R with cumulative distribution function $F_R(r)$, the **quantile** at level $\alpha \in (0, 1)$ is defined as:

$$q_\alpha = \inf\{r \in R : F_R(r) \geq \alpha\}.$$

Quantiles summarize the distribution by identifying threshold values below which a specified proportion of outcomes lie.

In the context of returns, lower quantiles correspond to extreme loss scenarios.

Value at Risk (VaR)

Definition

Value at Risk (VaR) is a risk measure defined in terms of the quantiles of the return distribution.

For a given confidence level α , the Value at Risk is defined as:

$$\text{VaR}_\alpha = -q_\alpha,$$

where q_α is the α -quantile of the return distribution.

Equivalently,

$$P(R \leq -\text{VaR}_\alpha) = \alpha.$$

In words, VaR_α represents the loss level that will not be exceeded with probability $1 - \alpha$ over a given time horizon.

Parametric VaR Under Normality

If returns are Normally distributed,

$$R \sim \mathcal{N}(\mu, \sigma^2),$$

then the α -quantile is given by:

$$q_\alpha = \mu + \sigma z_\alpha,$$

where z_α denotes the α -quantile of the standard Normal distribution.

Hence, the parametric VaR under the Normal assumption is:

$$\text{VaR}_\alpha = -(\mu + \sigma z_\alpha).$$

This closed-form expression is one of the main reasons the Normal model remains popular in practice.

Interpretation and Limitations

While VaR provides a simple and intuitive summary of risk, it has important limitations:

- It provides no information about the magnitude of losses beyond the VaR threshold.
- Under the Normal assumption, extreme tail risk is often underestimated.
- VaR is not a coherent risk measure in general.

These limitations motivate the study of alternative risk measures and more flexible distributional assumptions in subsequent modules.

Conclusion

In this module, the Normal distribution was introduced as a foundational model for asset returns. This assumption leads naturally to a quantile-based definition of risk through Value at Risk.

Although simplistic, this framework provides a mathematically clean starting point for understanding asset price dynamics and risk measurement. More refined models will build upon—and challenge—the assumptions made here.

Introduction

In the previous module, probability distributions were used to model asset returns and quantify risk. In this module, probability plays a fundamentally different role: it becomes the primary tool for assigning value to derivative securities.

An option is a financial contract whose payoff depends on the realization of a future random variable. As a result, option pricing is inherently a problem of probability theory and expectation.

This module introduces the probabilistic structure underlying option pricing, without yet invoking arbitrage arguments or risk-neutral measures.

Options as Random Variables

Let S_t denote the price of an underlying asset at time t . A European call option with strike price K and maturity T has a payoff given by:

$$X = \max(S_T - K, 0).$$

The payoff X is a function of the random variable S_T , and is therefore itself a random variable.

This observation is central: option pricing is concerned with the valuation of random future payoffs.

Probabilistic Model for the Underlying Asset

Assume that the underlying asset price evolves according to a probabilistic model. A common assumption is that log-returns are Normally distributed:

$$R_T = \ln \left(\frac{S_T}{S_0} \right) \sim \mathcal{N}(\mu T, \sigma^2 T).$$

Under this assumption, the terminal asset price can be written as:

$$S_T = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right),$$

where $Z \sim \mathcal{N}(0, 1)$.

Consequently, S_T follows a lognormal distribution.

Distribution of Option Payoffs

The option payoff is a nonlinear transformation of the terminal asset price:

$$X = g(S_T) = \max(S_T - K, 0).$$

Even if S_T follows a lognormal distribution, the distribution of X is not lognormal or Normal. Instead, it is induced by the distribution of S_T through the payoff function $g(\cdot)$.

This highlights an important feature of options: nonlinear payoffs distort the distribution of outcomes and concentrate probability mass in the tails.

Expected Payoff

A natural probabilistic quantity associated with the option payoff is its expected value:

$$E[X] = E[\max(S_T - K, 0)].$$

This expectation integrates the payoff over all possible realizations of the underlying asset price, weighted by their probabilities.

At this stage, the expectation is computed under an assumed probability measure, without yet specifying how future payoffs are converted into present values.

Discrete-Time Intuition: One-Period Binomial Model

To build intuition, consider a one-period binomial model for the asset price:

$$S_T = \begin{cases} S_0 u, & \text{with probability } p, \\ S_0 d, & \text{with probability } 1 - p, \end{cases}$$

where $u > d > 0$

The corresponding call option payoff is:

$$X = \begin{cases} \max(S_0 u - K, 0), \\ \max(S_0 d - K, 0). \end{cases}$$

The expected payoff is therefore:

$$E[X] = p \max(S_0 u - K, 0) + (1 - p) \max(S_0 d - K, 0).$$

This discrete example demonstrates that option pricing begins with the computation of expected payoffs under a specified probability distribution.

Role of the Probability Measure

The choice of probability measure plays a crucial role in option pricing. Different probability measures lead to different expectations and, consequently, different option values.

- A physical probability measure describes real-world asset behavior.

- A pricing probability measure is chosen to ensure consistency with market prices and the absence of arbitrage.

At this stage, it is sufficient to recognize that option prices are expectations, but not necessarily expectations under real-world probabilities.

Key Insight

Option pricing can be viewed as the evaluation of the expected value of a non-linear function of a random variable, where both the payoff structure and the chosen probability measure are essential.

This probabilistic perspective forms the foundation upon which no-arbitrage pricing and risk-neutral valuation will be developed in subsequent modules.

Conclusion

In this module, probability was introduced as the core mathematical framework for option pricing. By modeling the underlying asset probabilistically and treating option payoffs as random variables, option valuation naturally reduces to the computation of expectations.

This perspective sets the stage for the formal development of pricing measures, discounting, and continuous-time option pricing models.